

CLASS-PRIOR CONSTRAINTS ON CONFIDENT-TAIL TEST-TIME ADAPTATION FOR AUDIO DEEPPAKE DETECTION

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ABSTRACT

Confident-tail self-training assigns pseudo-labels to the lowest and highest scoring target examples. We study a limitation of this rule in audio deepfake detection: a selected class- c bucket occupying fraction b_c of a pool with class prior π_c has purity at most $\min(1, \pi_c/b_c)$, regardless of the detector. For exact symmetric tails of size q , perfect purity therefore requires $q \leq \min(\pi, 1 - \pi)$; this is not a guarantee of successful adaptation. On 400,435 available ASVspoof2021-DF eval trials, three released checkpoints fall from 98.78–99.11% accuracy to 61.80–65.24% after adaptation with nominal $q = 0.3$. Allocating the tails using a black-box label-shift estimate restores accuracy to 98.74–98.98%, while improvements over the source vary by checkpoint and metric. Controlled prevalence resampling reproduces the large budget’s degradation on In-the-Wild. Smaller tails avoid that large collapse without estimating a prior, whereas prior estimation itself can fail under corpus shift. Recomputed tie-aware metrics, threshold-only controls, and disjoint evaluation distinguish damage prevention from improved discrimination.

Index Terms— audio deepfake detection, test-time adaptation, label shift, pseudo-labels

1. INTRODUCTION

Self-supervised speech representations paired with anti-spoofing back-ends provide strong audio deepfake detectors [1, 2], but transfer across recording conditions and synthesis systems remains difficult [3]. A target-domain failure can involve both score ordering and the decision threshold. For example, a released detector in our In-the-Wild evaluation has ROC-AUC 0.9171 but only 39.08% accuracy at threshold 0.5. This motivates examining adaptation and re-thresholding separately; it does not establish that ranking always transfers or that probabilities are calibrated.

Test-time adaptation (TTA) uses unlabeled target observations to update a source model. Entropy minimization [4] and pseudo-label self-training can reinforce incorrect predictions. We examine a specific self-training rule: assign the bottom and top score quantiles to bona fide and spoof classes.

With a fixed symmetric budget, the minority bucket can be much larger than the entire minority population. Even a perfect ranking cannot fill that bucket with correct pseudo-labels. This obstruction is especially relevant when benchmark pools and deployment pools have different class proportions.

Class imbalance in adaptation is already established: Label Shift Adapter [5] estimates target proportions to adapt model parameters, and RLSbench [6] studies adaptation under joint label and conditional shift. Black-box shift estimation (BBSE) [7] estimates target priors from a source confusion matrix. Our contribution is narrower: an explicit counting constraint for confident-tail sampling, an audio checkpoint study of its consequences, and a comparison of prior-based allocation with smaller tails and threshold-only controls. T²A [8] instead addresses deepfake TTA through uncertainty-aware negative learning. We do not claim a new general label-shift estimator or superiority to specialized label-shift adaptation.

2. TAIL SELECTION AND ADAPTATION

Let $y = 1$ denote spoof and $s_\theta(x) \in [0, 1]$ the detector’s spoof score. The unlabeled adaptation pool has N clips and unknown class fractions π_c . Source labels are available for BBSE; target labels are used only for evaluation, controlled resampling, and explicitly marked oracle analyses.

2.1. A necessary condition for pure pseudo-labels

A bucket C_c assigned pseudo-label c , with realized fraction $b_c = |C_c|/N > 0$, contains at most $\pi_c N$ correct examples. Thus

$$\text{purity}(C_c) \leq \min\{1, \pi_c/b_c\}. \quad (1)$$

This is a finite-pool counting bound and makes no assumption about the model. For exact symmetric tails, $b_0 = b_1 = q$, perfect purity is possible only if

$$q \leq \min\{\pi, 1 - \pi\}, \quad \pi = \pi_1. \quad (2)$$

For illustration, at 97.22% spoof prevalence a bucket labeling 30% of the pool bona fide has purity at most 9.27%. High

AUC cannot remove this constraint. Satisfying it only removes the counting obstruction: overlapping score distributions can still contaminate both tails, and pure tails do not guarantee beneficial gradient updates. Conversely, learning may tolerate impure labels.

A prior allocation uses nominal fractions $q_c = 2q\hat{\pi}_c$, keeping the nominal total selection fraction $2q$. With exact bucket sizes and a correct prior, $q \leq 1/2$ ensures $q_c \leq \pi_c$. With estimated priors, the corresponding feasibility condition is

$$\hat{\pi}_c/\pi_c \leq 1/(2q) \quad (c = 0, 1; \pi_c > 0). \quad (3)$$

It limits permissible overestimation of each class; it is not a safety theorem. A smaller symmetric q is an alternative requiring no point estimate of the prior, but choosing a guaranteed feasible nonzero value still requires a lower bound on the minority prevalence.

Quantile ties. Our cached experiments use inclusive score thresholds, $s \leq Q(q_0)$ and $s \geq Q(1 - q_1)$, with spoof assignments taking precedence if masks overlap. Consequently, realized sizes can differ from nominal budgets. Equation (1) applies to the realized b_c ; equations (2)–(3) describe exact-size tails. We audit realized selections on cached source scores and do not interpret nominal budgets as exact sample counts. Deterministic selection of fixed counts would be a different implementation requiring new adaptation runs.

2.2. Prior estimation and objective

For $M_{ij} = P_S(\hat{y} = j \mid y = i)$ and target prediction histogram h , BBSE solves $M^\top \hat{p} = h$. We estimate M using up to 4,000 class-balanced clips from the checkpoint’s labeled training corpus; h uses source predictions on the adaptation pool. We clip the solution components to $[10^{-4}, 1 - 10^{-4}]$ and renormalize. The implementation returns a uniform prior if the matrix is singular. Because M uses training rather than held-out source data, optimism in its estimated error rates is a limitation.

At each of four passes, the current model scores the adaptation pool and refreshes pseudo-labels. For a mini-batch B and its selected subset B_C ,

$$\begin{aligned} \mathcal{L}_B = & \frac{1}{|B_C|} \sum_{x \in B_C} \text{CE}(p_\theta(x), \tilde{y}_x) \\ & + \frac{\lambda}{2|B|} \sum_{x \in B} \|p_\theta(a(x)) - \text{sg}[p_\theta(x)]\|_2^2, \end{aligned} \quad (4)$$

where $p_\theta = (1 - s_\theta, s_\theta)$, the CE term is omitted when B_C is empty, and $\lambda = 0.3$. The consistency term uses all batch examples and both probability components. The perturbation is per-clip gain sampled uniformly from $[0.7, 1.3]$ plus Gaussian noise with standard deviation 0.005. Only the last four transformer blocks’ LayerNorm affine parameters and the classifier head are updated; BatchNorm statistics stay frozen. Adam uses learning rate 10^{-4} and batch size 32. The released

DeepFense models have 16,706 trainable parameters under this rule. A fixed-confidence control instead selects scores at most 0.05 or at least 0.95, with the same objective.

3. EVALUATION PROTOCOL

Checkpoints and pools. We use DeepFense XLS-R 300M + AASIST checkpoints trained on ASVspoof2019-LA with training seeds 2, 42, and 240 [9, 10], denoted DF-2, DF-42, and DF-240. A WaveFake-trained SSL-AASIST checkpoint, WF [11], is evaluated on In-the-Wild. Audio is 16 kHz with the checkpoint-specific input length: 64,000 samples for DeepFense and 64,600 for WF.

The official ASVspoof2021-DF metadata [12] contains 611,829 trials across three phases, comprising 533,928 eval trials, 59,325 progress trials, and 18,576 hidden trials. Our available *eval-only* subset contains 400,435 trials (75.00% coverage): 389,299 spoof and 11,136 bona fide, or 97.219% spoof. Coverage by class is 75.00% spoof and 74.89% bona fide. The missing shard prevents a full-set comparison; these proportions do not establish an unbiased estimate of full-set EER. All method comparisons use the same available trials.

The main run adapts on 20,000 randomly selected clips with adaptation seed 0. We report both the available pool, which includes adaptation clips, and its 380,435-clip disjoint remainder. Additional adaptation seeds 1 and 2 use a fixed 50,000-trial subset, selected with seed 12345, with 20,000 clips for adaptation per run. These are sensitivity checks on overlapping data, not nine independent datasets. In-the-Wild uses all 31,779 clips (11,816 spoof; 37.182%), with 20,000 adaptation clips and an 11,779-clip disjoint remainder.

Prevalence intervention. On In-the-Wild with WF, we retain all spoof clips and randomly thin bona fide clips to spoof proportions 0.90, 0.95, and 0.97 (sampling seed 4242). These pools are smaller than 20,000, so adaptation and evaluation use the entire resampled pool. The experiment changes both prevalence and pool size; it is transductive and does not isolate prevalence from sample-count effects.

Metrics and controls. We recompute metrics from saved float32 probabilities. ROC-AUC assigns half credit to ties; EER linearly interpolates the crossing of FPR and FNR on the ROC, grouping identical scores. Interpolated EER need not correspond to an attainable deterministic threshold. We also report accuracy, balanced accuracy, and

$$G_\tau = \max_t \text{Acc}(s, t) - \text{Acc}(s, \tau). \quad (5)$$

The maximum considers all distinct score thresholds, including constant predictions. It uses evaluation labels and is strictly a diagnostic of threshold headroom, not a deployable method or a probability-calibration error. Unlike $1 - \text{EER}$, it is the actual attainable accuracy maximum on the observed scores.

Two label-free controls keep source scores fixed and select a threshold from adaptation scores: their median, or

Table 1. Recomputed results on the 400,435 available DF eval trials, checkpoint DF-2, adaptation seed 0. Acc. and balanced accuracy (BA) use threshold 0.5 except for the two frozen-source threshold controls. G_τ is an oracle diagnostic in percentage points; smaller is better. Baselines are the specified variants in Sec. 3.

Method	EER (%)	AUC	Acc. (%)	BA (%)	G_τ
Source	4.51	0.9923	98.87	88.28	0.20
Entropy (Tent-style)	4.28	0.9765	98.52	73.51	0.57
ETA reliability ablation	12.31	0.9288	98.93	82.71	0.03
SAR with scaled recovery	4.48	0.9923	98.88	88.10	0.17
IM-PL (SHOT-inspired)	6.35	0.9725	77.64	87.95	19.72
Fixed confidence 0.95	4.30	0.9918	99.02	85.03	0.00
Symmetric $q = 0.02$	4.19	0.9930	98.96	83.88	0.01
Symmetric $q = 0.3$	5.84	0.9846	62.96	80.89	35.38
BBSE tail budget	4.06	0.9931	98.95	82.34	0.02
Source + median threshold	4.51	0.9923	59.06	78.94	40.01
Source + BBSE quantile threshold	4.51	0.9923	98.88	88.19	0.19

their $(1 - \hat{\pi})$ -quantile using BBSE. The latter is a predicted-prevalence matching rule, not a guarantee of Bayes optimality. These controls preserve AUC and EER. Their cached-score prior estimates can differ slightly from the original adaptation-time forward passes.

We additionally report fixed-configuration entropy adaptation (Tent-style), an ETA reliability-only ablation without redundancy or Fisher regularization, and SAR with recovery threshold scaled to $(0.2/\ln 1000) \ln 2$ [4, 13, 14]. These variants share the above update settings. IM-PL is a SHOT-inspired information-maximization control with probability-space centroid pseudo-labels, not the original feature-space SHOT algorithm [15]; it freezes the head and updates 16,384 parameters. Thus these are controlled variants, not tuned reproductions establishing superiority over the published methods.

4. RESULTS

4.1. Preventing large-budget damage on DF

Table 1 shows that symmetric $q = 0.3$ lowers DF-2 accuracy from 98.87% to 62.96% and increases EER from 4.51% to 5.84%. Its oracle threshold gap is 35.38 points: much of the fixed-threshold damage is recoverable by re-thresholding, although AUC also declines. The BBSE budget reaches 98.95% accuracy and 4.06% EER. Fixed confidence reaches higher accuracy (99.02%), and the smaller $q = 0.02$ reaches 98.96% accuracy and 4.19% EER. Prior allocation is therefore one effective remedy, not a uniformly best choice. Balanced accuracy is lower after BBSE adaptation than for the source (82.34% versus 88.28%); the slight overall accuracy gain should not be interpreted as an improvement for both classes.

The accuracy collapse and its reversal occur on all three checkpoints (Table 2). Across the nine checkpoint/run combinations, BBSE has lower EER than the symmetric budget in 9/9 and the source in 8/9. We give these as descriptive counts without treating reused checkpoints and trials as independent

Table 2. DF checkpoint comparison on the same available eval pool. Entries are EER%/accuracy%; symmetric and BBSE budgets use nominal $q = 0.3$.

Checkpoint	Source	Sym.	BBSE
DF-2	4.51/98.87	5.84/62.96	4.06/98.95
DF-42	4.56/98.78	4.10/61.80	3.30/98.98
DF-240	3.82/99.11	4.66/65.24	3.80/98.74

evidence. The disjoint evaluation is more qualified: BBSE changes source EER from 4.52 to 4.07 (DF-2), 4.54 to 3.29 (DF-42), and 3.83 to 3.83 (DF-240, a 0.003-point increase before rounding). Avoiding severe damage is more consistent than improving an already strong source.

The frozen-source BBSE threshold gives 98.88% accuracy for DF-2, close to its 98.87% source result, while the median gives 59.06%. Median thresholding implicitly targets roughly balanced predictions and is poorly suited to this pool. The budget change can improve EER, which threshold changes cannot do, but it is not required to preserve high accuracy here. Removing consistency from the BBSE objective gives 3.98% EER versus 4.06% with it on DF-2; this single ablation provides no evidence that consistency is needed for the gain.

4.2. Score resolution and baseline interpretation

Tent-style entropy adaptation produces 98.07% of scores within 10^{-6} of 1.0, but only 93.02% exactly equal 1.0. ETA’s corresponding fractions are 97.71% and 96.65%. Conflating these quantities would make the stated AUCs appear impossible. With exact ties grouped, Tent-style adaptation has EER 4.28% and AUC 0.9765, compared with source 4.51% and 0.9923: a small improvement near one ROC operating point can coexist with worse integrated ranking. For ETA, ROC interpolation gives EER 12.31%; selecting the nearest ROC point and averaging its errors would give 7.44%, illustrating sensitivity to the scoring convention. For DF-42 and DF-240

Table 3. Native In-the-Wild, EER%/accuracy%, with nominal $q = 0.3$. WF is the WaveFake-trained checkpoint; all other rows use ASVspoof2019-trained models.

Checkpoint	Source	Sym.	BBSE
DF-2	9.64/88.30	9.71/85.69	9.89/87.13
DF-42	16.58/39.08	12.58/82.39	14.54/47.99
DF-240	8.47/68.92	8.33/88.80	8.48/76.98
WF	5.06/78.24	4.55/95.38	4.54/95.20

its balanced accuracy is 50%, indicating an all-spoof decision at threshold 0.5. These observations support reporting several metrics, not dismissing EER as arbitrary tie-breaking.

4.3. Corpus shift and prior-estimation failure

At the native In-the-Wild prior, the nominal symmetric budget has no counting obstruction. It improves accuracy on three checkpoints but reduces DF-2 accuracy (Table 3), consistent with feasibility being insufficient for a benefit. For DF-42, symmetric adaptation raises AUC from 0.9171 to 0.9473 and accuracy from 39.08% to 82.39%. A frozen-source median threshold already reaches 78.10%, showing how much the comparison depends on the operating point. The oracle threshold gap falls from 48.28 to 8.34 points.

BBSE instead estimates approximately 97.9% spoof on this 37.2%-spoof corpus and obtains 14.54% EER and 47.99% accuracy. A two-component score-mixture estimate gives 14.91%/55.00%. An oracle budget using the adaptation pool’s true prior gives 11.59%/91.28%. This isolates a substantial estimation cost on this checkpoint; the oracle is not a deployable method and is not always best on other checkpoints (WF: 4.67% oracle EER versus 4.55% symmetric).

BBSE is identifiable under label shift with an invertible confusion matrix, even for uncalibrated predictors [7]. Under unrestricted conditional shift, the hard prediction rate instead satisfies

$$P_{\mathcal{T}}(\hat{y} = 1) = \pi \text{TPR}_{\mathcal{T}} + (1 - \pi) \text{FPR}_{\mathcal{T}}, \quad (6)$$

which alone cannot determine all three unknowns. This explains why a source confusion matrix may fail to transfer. It does not prove impossibility for all output-based estimators with additional assumptions or side information, nor equivalence between prior error and calibration error.

4.4. Intervening on prevalence

The large-budget degradation also occurs on resampled In-the-Wild (Table 4). At 97% spoof, nominal $q = 0.3$ permits at most 10% purity in an exact-size bona fide tail and yields 65.42% accuracy; $q = 0.02$ yields 98.59%. However, the unchanged source reaches 98.94% there, so small q prevents the large collapse without universally improving the source. The experiment supports prevalence sensitivity of this selection

Table 4. WF on resampled In-the-Wild: accuracy% for the unchanged source and symmetric-tail adaptation. Pool size varies with prevalence.

π	N	Source	$q = .02$	$q = .10$	$q = .30$
0.90	13,129	96.49	97.75	95.93	73.27
0.95	12,438	98.30	98.60	92.64	68.48
0.97	12,181	98.94	98.59	89.39	65.42

rule; it neither proves inevitable failure whenever the bound is below one nor identifies prevalence as the only cause of adaptation failure.

5. DISCUSSION AND LIMITATIONS

The counting obstruction gives a useful diagnostic for pseudo-label allocation, while its consequences depend on score quality, realized selection sizes, and the learning objective. Recomputed metrics support severe fixed-threshold damage from large symmetric tails and its mitigation by prior allocation or smaller tails. They also show important tradeoffs: some remedies reduce minority-class performance, prior estimates can fail on shifted corpora, and re-thresholding alone explains a substantial part of some accuracy gains.

The experiments use four released checkpoints from one broad architecture family, two principal corpora, an incomplete DF eval partition, and fixed adaptation hyperparameters. They do not establish general superiority over specialized label-shift methods. Our prevalence intervention changes sample count, and the seed checks share data. Future validation should use independent corpora, fixed-size prevalence interventions, held-out source calibration, exact-count tail selection, and development-domain tuning of baseline methods. Neither a small nominal tail nor source-score agreement alone certifies safe adaptation without assumptions about the target distribution.

Reproducibility. The script `audit_icassp.py` verifies 89 saved score arrays against historical AUC and accuracy, recomputes the reported metrics, and writes the tables and input hashes. The project repository is <https://github.com/RohamIzadidoost/Kaggle-deepFakeAudio>. The accompanying audit bundle includes the revised scripts and outputs. No new neural-network training is required to reproduce these tables from the saved scores and original manifests.

Ethics and assistance. This study uses existing public datasets and checkpoints; no new human-subject data were collected. OpenAI Codex assisted with the revised manuscript text, metric-audit code, and result tables; the adaptation experiments predate this revision.

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